

PROVISIONAL PATENT APPLICATION

MODULAR STACKABLE FASTENER-SORTING AND STORAGE TRAY SYSTEM

Inventor: Timothy R. Williams

Residence: Carson City, Nevada

Draft Date: August 20, 2026

This document has not been filed with the United States Patent and Trademark Office.

TITLE OF THE INVENTION

Modular Stackable Fastener-Sorting and Storage Tray System

FIELD OF THE INVENTION

The present disclosure relates generally to storage and organization devices and, more particularly, to modular trays configured to sort, retain, identify, display, and stack elongated fasteners or similar small components according to size, type, material, application, or another selected classification.

BACKGROUND OF THE INVENTION

Loose screws and other elongated fasteners are commonly stored in jars, bags, bins, compartment boxes, or original packaging. These storage methods can allow multiple sizes or types to become mixed, require the user to search through a container, and make it difficult to compare a loose fastener with labeled stock. If a container is dropped or tipped, its contents may scatter or become combined with other hardware.

Conventional compartment organizers provide separated spaces but generally conceal most of each fastener, occupy a fixed volume regardless of inventory, and do not necessarily provide an integrated measuring, labeling, or display function. A need therefore exists for a compact, modular system that visibly separates elongated fasteners, identifies the stored category, retains the items in organized channels, and permits multiple trays to be stacked or rearranged without requiring one large permanent cabinet or box.

SUMMARY OF THE INVENTION

The invention provides a modular fastener-storage tray having a generally planar body, a plurality of elongated retention channels, an identification region, and connector features located near peripheral portions of the tray. Each retention channel is configured to receive one or more screws or other elongated articles while keeping the articles visible and separated from articles placed in adjacent channels.

In one embodiment, corner connector structures include upward and downward engagement features that permit two or more trays to be stacked with a vertical spacing between adjacent tray bodies. The trays may differ in width, height, number of channels, channel geometry, and identification markings while retaining compatible connector locations. A user may therefore build a storage stack suited to the available fastener inventory and may remove, replace, or reorder an individual tray without replacing the entire system.

The identification region may include permanently formed or removable labels indicating a fastener dimension, material, thread designation, use category, owner, location, or other information.

Example labels include PLA, PETG, ABS, TPU, UAT, M2, M3, and M3 x 4. The system may be produced by additive manufacturing as a unitary tray or may be molded, machined, cut, assembled, or manufactured by another suitable process.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a top engineering view of three example Screw Stack tray configurations.

FIG. 2 is a side engineering view showing example tray thicknesses and connector projections.

FIG. 3 is a bottom engineering view showing the underside geometry and connector locations.

FIG. 4 is a perspective CAD rendering of an example labeled tray having elongated retention channels.

FIG. 5 is a perspective multi-view engineering drawing of example tray configurations.

FIG. 6 is a photograph of a manufactured prototype with example screws positioned in the retention channels.

DETAILED DESCRIPTION OF THE INVENTION

A modular fastener-storage system includes one or more trays. Each tray has a generally planar body extending between a top surface and a bottom surface. The illustrated tray bodies are generally rectangular, although square, circular, polygonal, curved, or irregular tray shapes may be used. Example illustrated bodies have widths of approximately 2.6 inches, 3.0 inches, or 3.4 inches and heights of approximately 1.6 inches or 2.4 inches. These dimensions describe examples only and do not limit the system to a particular scale.

A plurality of elongated retention channels extends across a portion of the tray body. The channels may be open slots, recessed grooves, apertures, resilient clips, rails, or combinations thereof. In an illustrated embodiment, the channels are substantially parallel and sized to receive the shafts of screws while at least a portion of each screw head remains supported above or adjacent to the channel. Multiple fasteners may be arranged along a channel for visible counting, comparison, or retrieval.

The channels may have a common width or may include different widths on one tray. A tray may be dedicated to a selected fastener diameter, thread designation, length range, head style, material, or application. Channel ends may be closed to prevent a fastener from sliding off the tray, partially

open for side loading, or provided with a removable stop. Recesses, ramps, finger-clearance areas, or enlarged loading regions may assist with insertion and removal.

An identification region is positioned adjacent to the retention channels. The region may contain raised, recessed, printed, engraved, molded, replaceable, magnetic, adhesive, or electronically displayed information. The illustrated prototypes include a product name and separate label fields for material, organization, and fastener size. A blank or replaceable label panel may allow a user to revise the classification without replacing the tray.

Connector structures are positioned near peripheral portions of the tray. In the illustrated rectangular embodiments, connector structures are positioned near four corners. Each connector structure may include one or more sockets, projections, pegs, posts, rails, clips, magnets, threaded members, or friction-fit elements. Compatible upper and lower engagement features permit one tray to be supported above another while maintaining access to at least part of the lower tray.

Illustrated connector details include features approximately 0.08 inch and 0.10 inch in width and example tray or connector thicknesses of approximately 0.20 inch, 0.25 inch, 0.30 inch, or 0.35 inch. Peripheral connector regions may extend approximately 0.40 inch from the body. These dimensions are exemplary and may be changed according to fastener size, material, manufacturing tolerance, desired stack spacing, and anticipated load.

The modular connectors permit trays having different body sizes or storage purposes to be combined in one stack when their engagement features share a compatible spacing or adapter. Intermediate connector pieces may join trays, increase vertical clearance, offset adjacent trays, join trays side by side, or attach the system to a wall, drawer, toolbox, workbench, pegboard, mobile cart, or other support.

Prototype trays were designed in Fusion 360, prepared in slicing software, and produced by fused-filament fabrication. Early testing identified disconnected peg geometry, text that did not remain legible after slicing, and channel profiles that required deeper fillets or revised clearances.

Subsequent versions connected the engagement geometry, increased or simplified label features, and revised the channels to improve printing and fastener fit. These changes demonstrate that the tray geometry may be parametrically adapted to different printers, nozzles, materials, and fastener categories.

The tray may be formed from PLA, PETG, ABS, TPU, nylon, composite filament, metal, wood, or another material. A more flexible material may permit snap-fit retention or resilient channel walls,

while a more rigid material may improve stacking stability. The tray may be additively manufactured as one component or assembled from a body, label modules, channel inserts, and connector pieces. Alternative production methods include injection molding, CNC machining, laser cutting, stamping, casting, or thermoforming.

Optional embodiments may include a transparent cover, removable lid, drawer slide, handle, wall-mount feature, inventory scale, color coding, barcode, QR code, radio-frequency identification tag, near-field communication tag, sensor, illumination, or electronic inventory interface. These optional features are not required for operation of the basic mechanical tray system.

EXAMPLE METHOD OF USE

A user selects a tray associated with a fastener category, places screws or other elongated articles into corresponding retention channels, identifies the category in the tray's identification region, and connects the tray to one or more compatible trays. The user may visually inspect and retrieve an individual fastener while the remaining articles stay separated. When the storage requirement changes, the user may remove, replace, relabel, or reorder an individual tray.

CLAIMS

1. A modular storage tray for elongated fasteners, comprising: a tray body having a top surface and a bottom surface; a plurality of elongated retention channels formed in the tray body and configured to receive one or more elongated fasteners; an identification region associated with the plurality of elongated retention channels; and at least one connector structure configured to removably connect the tray body with another storage tray.
2. The modular storage tray of claim 1, wherein the plurality of elongated retention channels comprises substantially parallel slots extending across the tray body.
3. The modular storage tray of claim 1, wherein at least one retention channel is configured to receive a shaft portion of a screw while supporting a head portion of the screw adjacent to the top surface.
4. The modular storage tray of claim 1, wherein the identification region includes at least one raised, recessed, printed, engraved, molded, removable, or electronically displayed label.

5. The modular storage tray of claim 1, wherein the at least one connector structure comprises an upper engagement feature and a lower engagement feature configured to vertically stack the tray body with a second tray body.
6. The modular storage tray of claim 5, wherein the connector structures maintain a vertical clearance between adjacent stacked tray bodies.
7. The modular storage tray of claim 1, wherein connector structures are positioned adjacent to a plurality of corners of the tray body.
8. The modular storage tray of claim 1, wherein the tray body, retention channels, identification region, and connector structure are formed as a unitary additively manufactured component.
9. The modular storage tray of claim 1, further comprising at least one removable channel insert, label insert, stop, spacer, cover, or mounting adapter.
10. A modular storage system comprising: a first storage tray according to claim 1; a second storage tray having a different tray-body dimension, channel dimension, channel count, or identification marking; and compatible connector structures removably joining the first storage tray and the second storage tray.
11. The modular storage system of claim 10, further comprising an adapter configured to join trays having different body dimensions or connector locations.
12. A method of organizing elongated fasteners, comprising: selecting a modular tray associated with a fastener category; placing one or more elongated fasteners into retention channels of the modular tray; identifying the fastener category within an identification region of the modular tray; and removably connecting the modular tray to at least one additional tray.

ABSTRACT

A modular storage tray is configured to sort, retain, identify, display, and stack elongated fasteners or similar articles. The tray includes a generally planar body, a plurality of elongated retention channels, an identification region, and connector structures for removably joining the tray with one or more additional trays. Different trays may provide different dimensions, channel counts, channel geometries, and labels while retaining compatible connector locations. The connectors may vertically space adjacent trays to preserve access to stored fasteners. The tray may be additively manufactured as a unitary component or produced by molding, machining, cutting, or assembly.

Optional embodiments may include replaceable labels, channel inserts, covers, mounting adapters, color coding, machine-readable identifiers, or electronic inventory features.

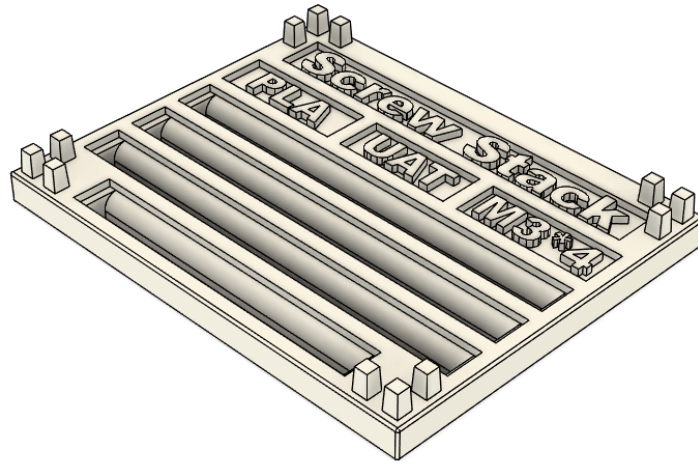


FIG. 1 - Perspective CAD Rendering of an Example Tray



FIG. 2 - Manufactured Prototype with Example Screws

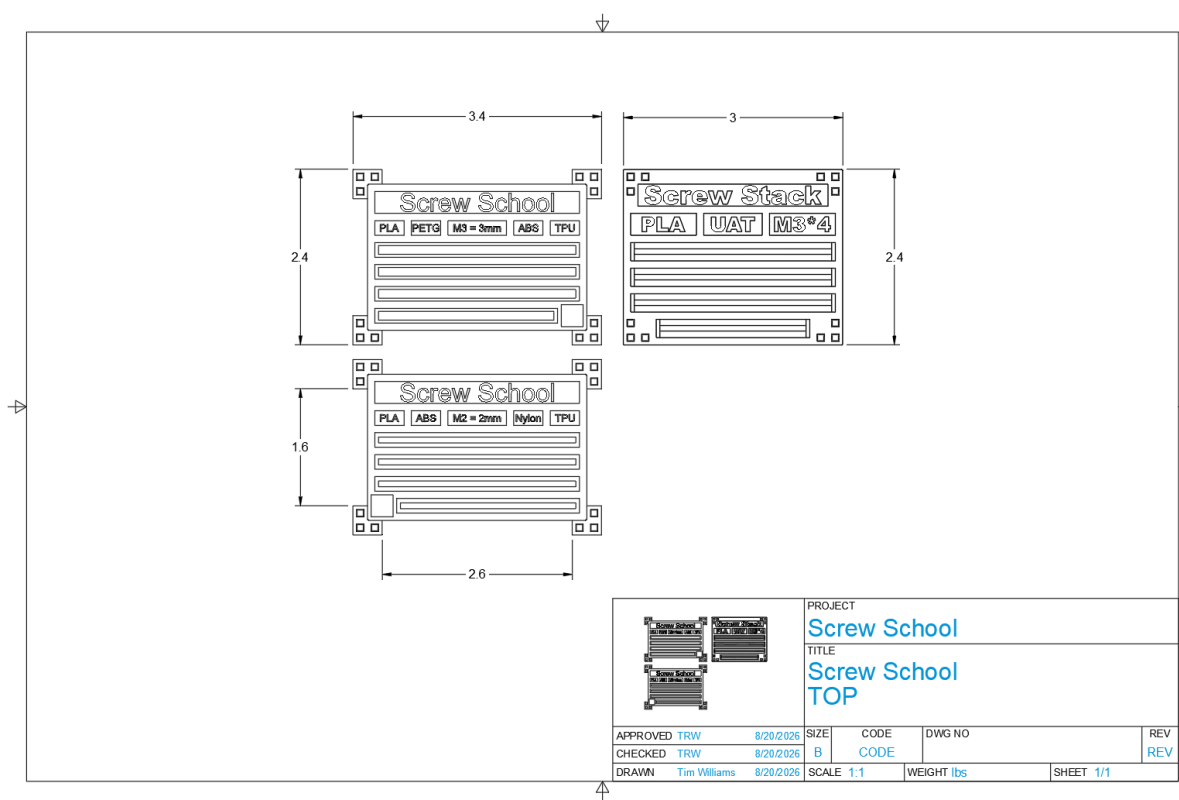


FIG. 3 - Top Views and Principal Tray Dimensions

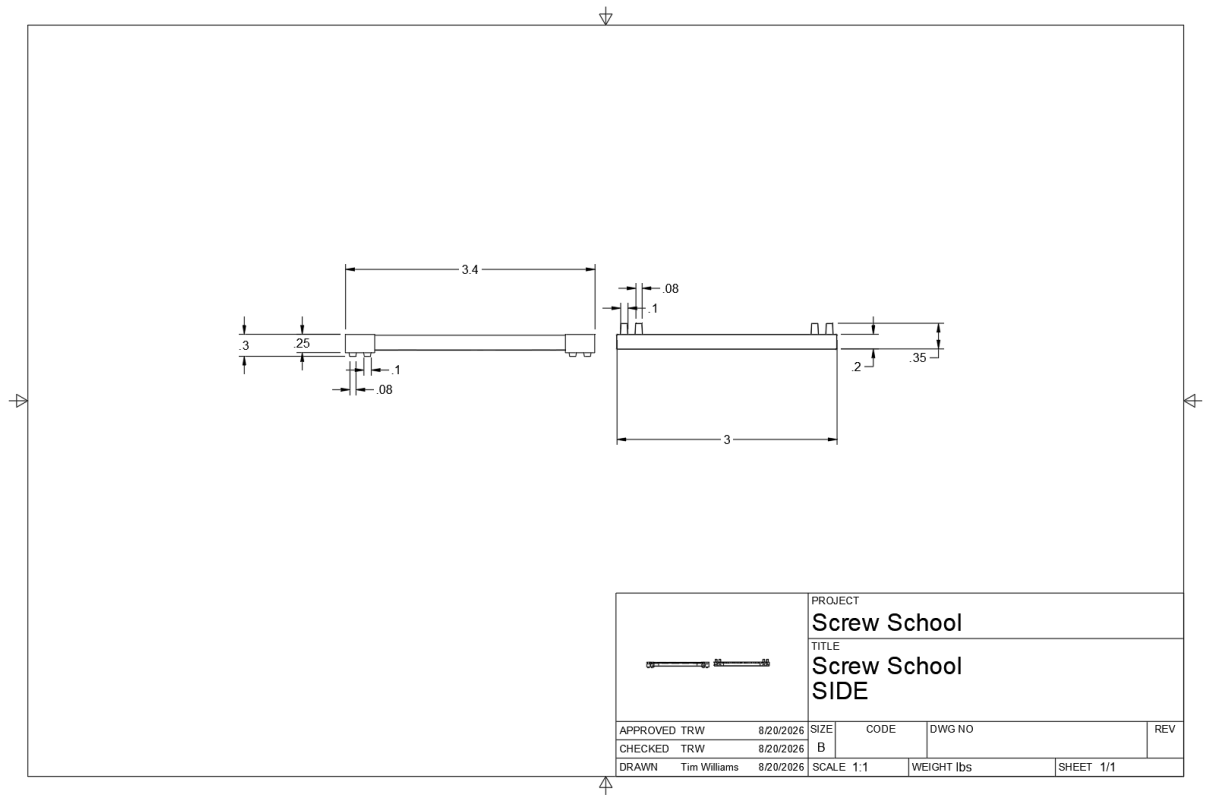


FIG. 4 - Side Views, Thicknesses, and Connector Projections

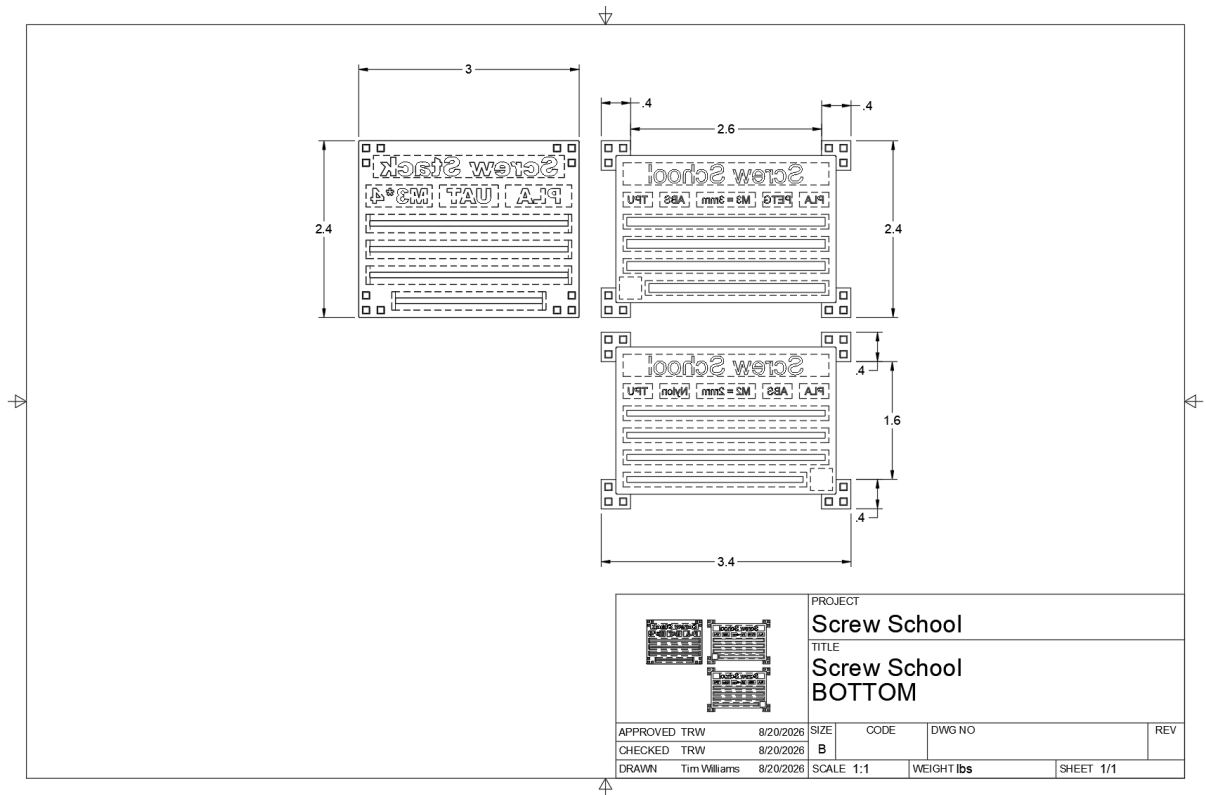


FIG. 5 - Bottom Views and Connector Locations